Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding for Augmented Reality (AR) Applications

TGS Ref No: TGS-2022017519

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 9.1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
8.0	4 July 2026	Legacy trainer-slide-aligned lesson plan.	Tertiary Infotech Academy
9.0	11 September 2026	Rebuilt for the AI vibe-coding AR/VR course, 215-slide deck, localhost WebXR studio, eight labs and aligned WA/PP.	Dr Alfred Ang
9.1	12 September 2026	Added the live AR/VR Playground as the no-install learner entry point.	Dr Alfred Ang

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Course Information

Course title	AI Vibe Coding for Augmented Reality (AR) Applications
Course code	TGS-2022017519
TSC	Augmented Reality Application · PRE-CTS-3001-1.1
Duration	2 days · 16 training hours
Daily timing	09:30–18:30, with 1-hour lunch excluded from the 8-hour training total
Mode	Instructor-led technical instruction, supervised localhost coding labs and device-dependent WebXR demonstrations
Provider	Tertiary Infotech Academy Pte Ltd · UEN 201200696W
Version	v9.1 · 12 September 2026

Learning Outcomes

- LO1: Translate specifications and write storyboards for AR applications using spatial user-interface principles.
- LO2: Apply AR working principles to create procedures and review AR runtimes for usability and clarity.
- LO3: Overlay virtual objects on the real world, distinguish AR from VR, and advise developers on effective adjustments.

Assessment

- Written Assessment (SAQ) — 7 open-ended questions, 60 minutes, open book.
- Practical Performance (PP) — 3 connected tasks, 90 minutes, open book.
- The current papers total 150 minutes; this conflicts with the generic 2-hour course-page label and is retained to mirror the approved paper-level evidence.

Facilitation and Evidence Strategy

- Slides present mechanisms, architecture, code/configuration excerpts, formulas, failure modes and acceptance evidence; they do not reproduce click-by-click lab procedures.
- The Learner Guide and each individual lab folder carry the complete numbered procedure, verification, troubleshooting and cleanup.
- The shared localhost app provides desktop preview in all supported modern browsers; immersive AR/VR demonstrations are described honestly as device-dependent.

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- Coding-agent output is reviewed as a proposal. Learners inspect the scene contract, diff, tests, runtime state and handover before acceptance.

Course Schedule

Day 1 — AR User Interface Design and Storyboarding with AI Vibe Coding

Time	Minutes	Type	Learning activity	Slide reference
09:30–10:00	30	Admin	Welcome, digital attendance, trainer profiles, ground rules, course download, outcomes and assessment overview	Slides 1–17
10:00–10:45	45	Topic	Topic 1 — AR User Interface Design and Storyboarding with AI Vibe Coding: requirement contracts, evidence boundaries and storyboard state models	Slide 18
10:45–11:00	15	Break	Tea break	—
11:00–12:00	60	Lab	Lab 01 — Scene Brief and Prompt Contract	Slides 34–37
12:00–13:15	75	Lab	Lab 02 — Storyboard and Spatial UI	Slides 53–56
13:15–14:15	60	Lunch	Lunch break	—
14:15–15:15	60	Lab	Lab 03 — WebXR Capability and Equipment Gate	Slides 72–75
15:15–15:30	15	Break	Tea break	—
15:30–16:30	60	Topic	Topic 2 — AR Application Development	Slide 80

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			with Localhost WebXR: renderer, scene graph, camera, lighting and metric scale	
16:30–18:00	90	Lab	Lab 04 — Scene Graph, Lighting and Animation	Slides 97–100
18:00–18:30	30	Lab	Lab 05 — Plane Detection, Hit Test and Anchors (brief, state trace and setup)	Slides 116–117

Training total excluding lunch: 480 minutes (8 hours).

Day 2 — Interaction, AR/VR Review and Assessment

Time	Minutes	Type	Learning activity	Slide reference
09:30–10:30	60	Lab	Lab 05 — Plane Detection, Hit Test and Anchors (runtime placement, verification and review)	Slides 118–119
10:30–10:45	15	Break	Tea break	—
10:45–12:00	75	Lab	Lab 06 — Raycast, Touch and Spatial Feedback	Slides 135–138
12:00–13:15	75	Lab	Lab 07 — Immersive VR, Controllers and Comfort with Topic 3 AR/VR runtime boundary and comfort mechanisms	Slides 165–168
13:15–14:15	60	Lunch	Lunch break	—
14:15–15:45	90	Lab	Lab 08 — Runtime	Slides 184–187

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			Review, Optimisation and Handover: evidence, bounded adjustment, regression and handover	
15:45–16:00	15	Break	Tea break and assessment attendance	—
16:00–17:00	60	Assessment	Written Assessment (SAQ) — 7 open-ended questions, open book	Closing assessment block
17:00–18:30	90	Assessment	Practical Performance — 3 connected tasks, open book	Closing assessment block

Training total excluding lunch: 480 minutes (8 hours).

Lab-to-Slide and TSC Alignment

Lab	Title	Minutes	TSC/LO codes	Deck
01	Scene Brief and Prompt Contract	60	LO1 · K3 · A1	Slides 34–37
02	Storyboard and Spatial UI	75	LO1 · K1 · K2 · A2	Slides 53–56
03	WebXR Capability and Equipment Gate	60	LO1 · K3 · K5	Slides 72–75
04	Scene Graph, Lighting and Animation	90	LO2 · K4 · K5	Slides 97–100
05	Plane Detection, Hit Test and Anchors	90	LO2 · K5 · K6	Slides 116–119
06	Raycast, Touch and Spatial Feedback	75	LO2 · K2 · A3	Slides 135–138
07	Immersive VR, Controllers and	75	LO3 · K7	Slides 165–168

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	Comfort			
08	Runtime Review, Optimisation and Handover	90	LO3 · A3 · A4	Slides 184–187

Resource and Safety Notes

- Live learner Playground: https://alfredang.github.io/arvr-playground/ (static HTML/CSS/JavaScript; no installation required).
- Learner laptop with a current Node.js LTS release and modern Chromium-based browser for source-editing labs.
- Optional supported AR/VR device for immersive evidence; desktop preview remains available.
- A clear physical play area for immersive demonstrations and active supervision.
- Synthetic/local assets only; credentials, personal data and trainer-only answer keys are excluded from coding-agent prompts and public repositories.
- Current source references are recorded in the Learner Guide.

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